MILITARY SPECIFICATION

CLOCK, AIRCRAFT, MECHANICAL ABU-32/A

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers an integrally lighted mechanical aircraft clock with a 12-minute elapsed time capability, designated ABU-32/A.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

DD-G-451	Glass, Flat and Corrugated, for Glazing, Mirrors, And Other Uses
PPP-T-360	Time Measuring Instruments: Packaging of
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-G-174	Glass, Optical
MIL-C-675	Coating of Glass Optical Elements (Antireflection)
MIL-R-6855	Rubber; Synthetic, Sheeted Molded, and Extruded
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-A-26482	Connectors, Electric, Circular, Miniature, Quick Disconnect
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for
MIL-B-27497	Bearing, Jewel, Sapphire, Synthetic
MIL-C-83723	Connector, Electrical # Circular Environment Resisting #, Receptacles and Plugs, General Specfications for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Engineering Division: San Antonio ALC/MMEDO, Kelly AFB, TX 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6645

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DOD-D-1000 Drawings, Engineering and Associated Lists

STANDARDS

FEDERAL

FED-STD 595 - Colors.

(Requirements for individual color chips, 3 x 5 Supplements should be submitted to the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120).

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-147	Palletized Unit Loads
MIL-STD-794	Parts and Equipment Procedures and Packaging and Packing of
MIL-STD-810	Environment Test Methods for Aerospace and Ground Equipment
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MS3132	Connector, Receptacle, Electric, Solder Type, Box Mounting Flange, No. 4 Holes, Push Pull Coupling
MS3134	Connector, Plug, Electric, Solder Type, Short, Push Pull Coupling
MS3137	Connector, Plug, Electric, Solder Type, Short, Push Pull Coupling
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS51021	Setscrew - Hexagon Socket, Cup Point, Corrosion-Resisting (Chrome-Nickel) Steel, Passivated, UNC-3A

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein.

American Society For Testing and Materials (ASTM)

ASTM D-3951 - Packing, Commercial (Application for copies should be addressed to American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19130)

2.2.1 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First Article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.4).

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3.2 Materials.

3.2.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity.

3.2.2 Antimagnetic materials. Antimagnetic materials shall be used for all parts of the clock, except where magnetic materials are essential. When the use of magnetic material is necessary, all magnetic material shall be demagnetized.

3.2.3 Metals. Metals shall be corrosion resistant or suitably treated to resist corrosion due to fuels, salt spray, or other atmospheric conditions likely to occur in storage or normal service. Protective coatings that will crack, chip, or scale with age or extremes of atmospheric conditions shall not be used.

3.2.3.1 Dissimilar Metals. Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact with each other.

3.2.3.2 Light metal alloys. Aluminum, magnesium, and other lightweight alloys shall be used for all metal parts of the clock, except where stressing, fire resistance, or other requirements dictate the use of steel or other heavy metals.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Design and construction. The clock shall be designed and sealed to ensure that sand, dust, humidity, salt air, and fungus producing conditions in service life will not interfere with the proper operation of the equipment. The clock shall be so constructed that no parts will work loose in service. The clock shall be built to withstand without failure, the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service use.

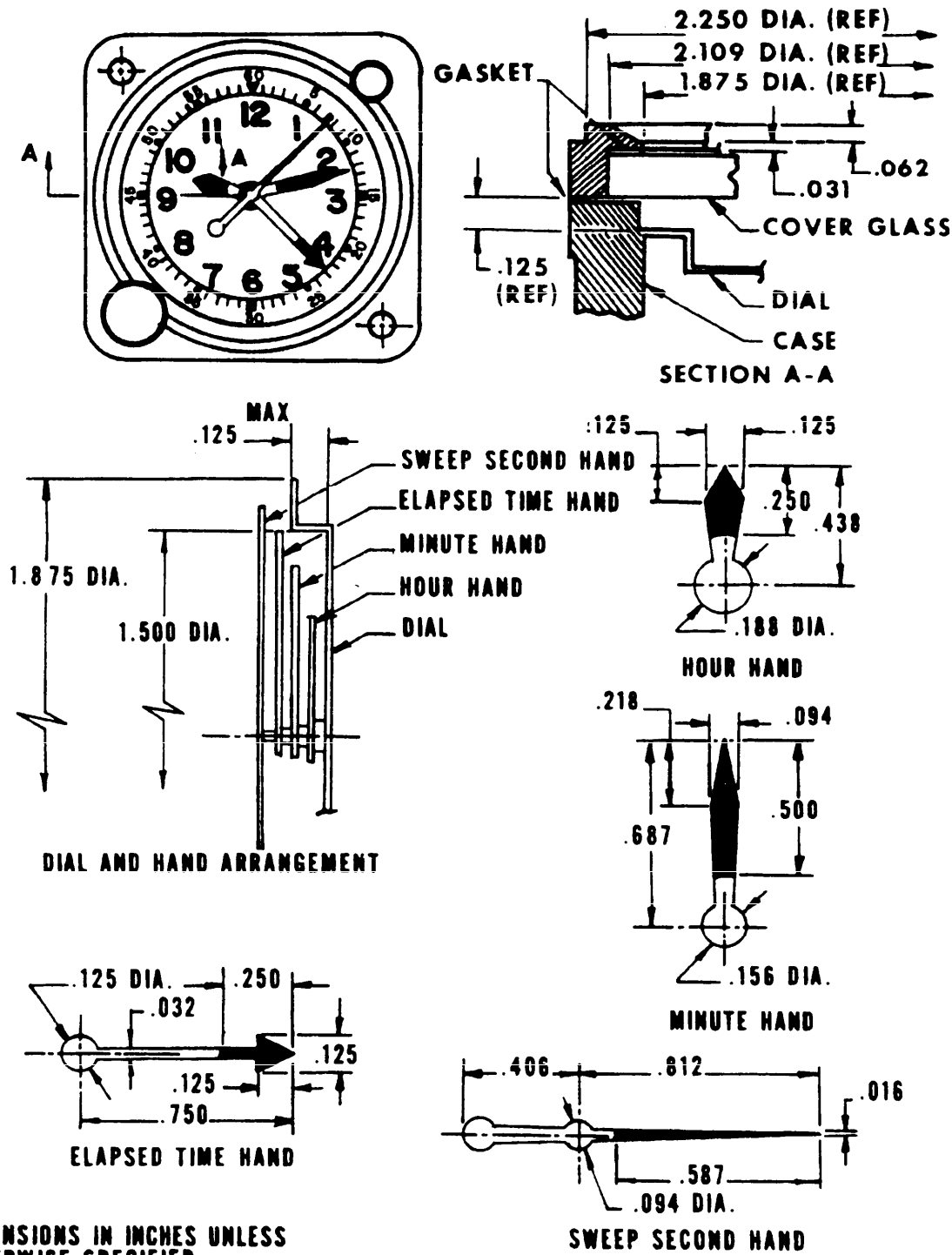
3.4.1 Clock case. The case must fit Aeroquip Corporation, 300 South East Avenue, Jackson, MI 49203-1972, instrument clamp, part number 52984, or equivalent.

3.4.1.2 Cover glass.

3.4.1.2.1 Dimensions. The diameter of the cover glass shall be 2.052 inches +0.077 or -0.010 inch. The thickness shall be 0.062 +0.016 or -0.000 inch, except as specified in 3.4.1.2.3, when applicable.

3.4.1.2.2 Material. The case cover glass shall conform to type II, quality AA of DD-G-451, except as specified in 3.4.1.2.3. Any flaws permitted by DD-G-451 shall not interfere with reading of the clock. The edges of the glass shall be ground or otherwise treated to remove roughness, nicks, or sharpness.

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DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS±.005

FIGURE 1. CLOCK HANDS

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3.4.1.2.3 Exception for prism wedge glass. If a prism wedge glass is used for better light distribution, the material for the cover glass shall be in accordance with MIL-G-174. Any flaws permitted by MIL-G-174 shall not interfere with reading the clock. The thin edge of each prism shall be not less than 0.0468 inch.

3.4.1.2.4 Coverglass and lighting wedge. All glass surfaces shall be provided with a reflection-reducing coating conforming to MIL-C-675, in addition to withstanding the environmental conditions specified herein, except that the following reflectance tolerances shall apply:

Angle of incidence	Wavelength - millimicrons	Percent reflectance
Zero degree	450 - 675	0.6 max.
Zero degree	425 - 700	0.5 average
30 degrees	450 - 625	1.0 max.
30 degrees	425 - 700	0.5 average

3.4.1.2.5 Cover glass gasket. The cover glass or bezel gasket shall conform to class 2, grade 60 of MIL-R-6855, or a nonshrinking material acceptable to the procuring acitivity.

3.4.1.3 Bezel. If a separate bezel is used, it shall be constructed from nonferrous, low-density metal, or plastic material, finished with a durable lusterless black, color No. 37038, in accordance with FED-STD 595. The bezel shall be held in place by at least four screws.

3.4.1.4 Threaded screw ring. The threaded screw ring, if used, shall be removable from the case without damaging the case or glass. Steel or magnetic metal shall not be used.

3.4.2 Hands and dial

3.4.2.1 Hands. The clock hands shall conform to the dimensions shown on figure 1, with a thickness equal to accepted commercial requirements. The hands shall be lightweight, sufficiently rigid to prevent oscillation under vibration, firmly attached to the mechanism, yet readily adjustable.

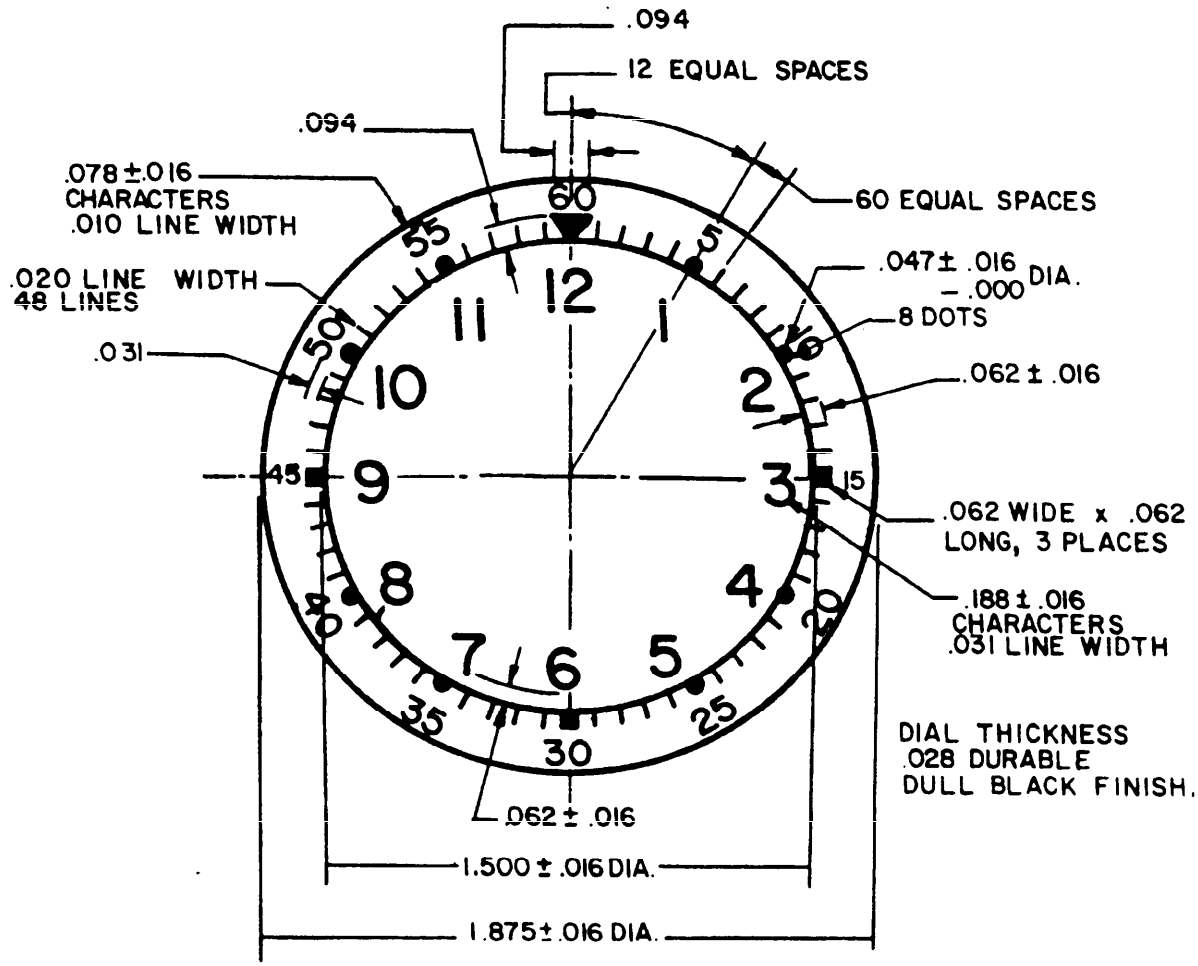
3.4.2.2 Dial. The clock dial shall be constructed from aluminum or aluminum alloy and shall have a step arrangement as shown on figure 1. The dial dimensions and markings shall conform to figure 2. The dial shall be securely fastened by at least two screws into the case or frame of the mechanism so it will not loosen or turn when the clock is vibrated.

3.4.2.2.1 Dial and cover glass location. The distance between the inside surface of the cover glass and the cover glass and the main portion of the dial shall not exceed 0.25 inch. The distance between the glass and the stepped up portion of the dial shall not exceed 0.125 inch.

3.4.2.3 Colors for dial and hands. The dial background and area of the hands unshaded on figure 1 shall be lusterless black conforming to color No. 37038 of FED-STD 595. The shaded area of the hands and the dial numbers and markings shall be lusterless white conforming to color No. 37875 of Federal Standard No. 595.

3.4.3 Clock controls.

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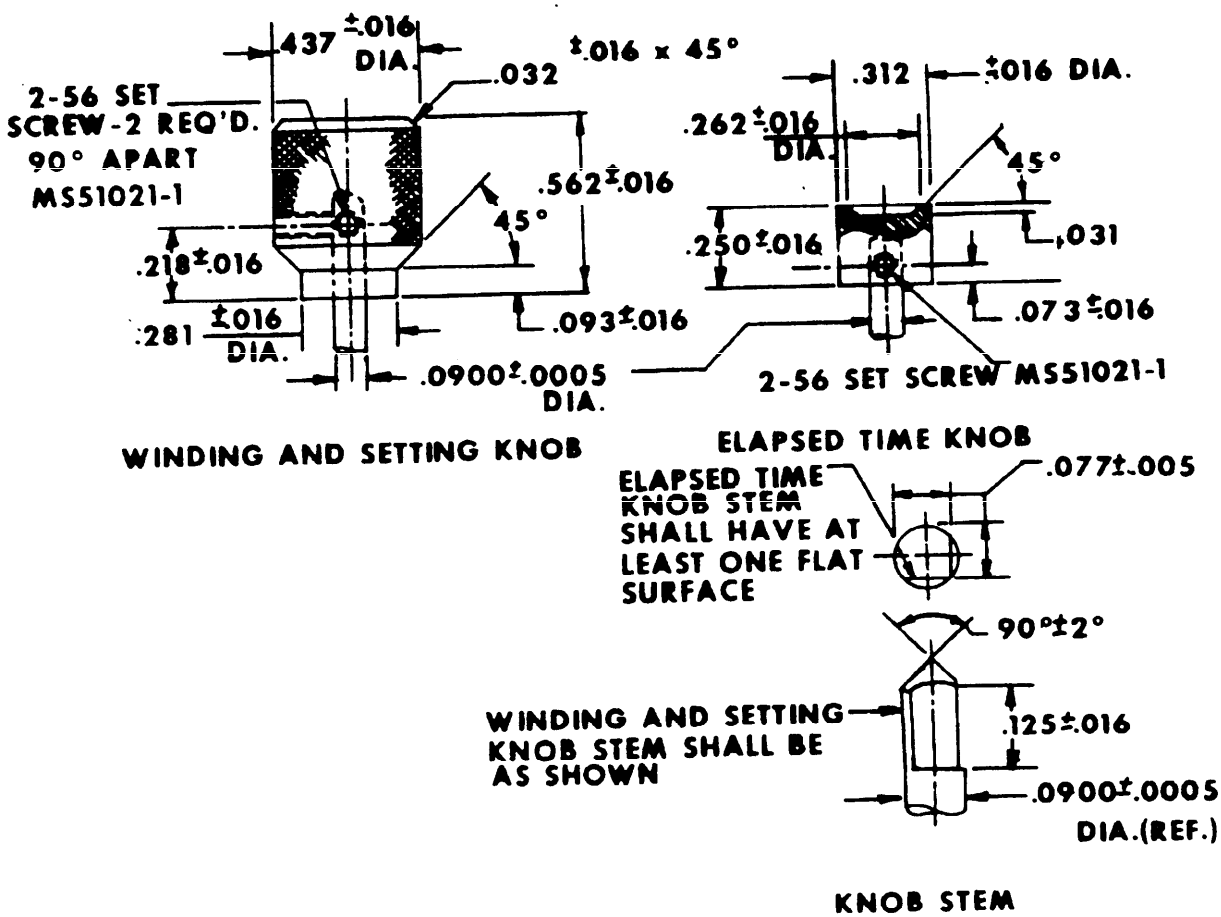


ENLARGED VIEW

DIMENSIONS IN INCHES
UNLESS OTHERWISE
SPECIFIED, TOLERANCES:
DECIMALS ± .005

FIGURE 2. CLOCK DIAL

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DIMENSIONS IN INCHES .
UNLESS OTHERWISE SPECIFIED.
TOLERANCES: DECIMALS .005

FIGURE 3 CLOCK KNOBS

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3.4.3.1 Winding and setting knob. A winding and setting knob conforming to the dimensions on figure 3 shall be located in the lower left-hand corner of the clock case. It shall wind the mainspring of the clock when turned in a clockwise direction and shall be free to turn counterclockwise with a ratchet gear. When pulled out, it shall engage the setting gear for the hour and minute hands of the clock into the setting position. The knob shall have a diamond knurl surface and shall be attached to the shaft with two socket head setscrews conforming to MS51021-1(2-56).

3.4.3.2 Elapsed time knob. An elapsed time knob conforming to the dimensions on figure 3 shall be located in the upper right-hand corner of the clock case to control the elapsed time function of the clock. Pushing the knob shall operate the three phases of this function as follows: When the elapsed time and sweep second hands are at rest at "60" on the dial, pushing the knob shall cause them to start; if the hands are moving, pushing the knob shall cause them to stop; if the hands are stopped elsewhere on the dial than at "60", pushing the knob shall cause them to fly back to "60". The elapsed time function shall not interfere with the hour and minute time hands. The knob shall be attached to the shaft with one socket head setscrew conforming to MS51021-1 (2-56).

3.4.3.3 Stems. The winding stem and the elapsed time stem shall be made of steel with a minimum carbon content of 0.75 percent and with a minimum degree of hardness equal to Rockwell C45. Each stem shall be milled to conform to dimensions shown on figure 4 to provide for "seating" the setscrews.

3.4.4 Balance wheel and hairspring unit. The balance wheel and hairspring unit shall be temperature compensated.

3.4.4.1 Balance staff. The balance staff shall be readily removable and replaceable without disturbing the poise of the balance wheel.

3.4.4.2 Hairspring. The hairspring shall be constructed from corrosion-resistant metal.

3.4.4.3 Escapement. A fully jeweled lever escapement, including a roller jewel and two pallet stones, shall be provided.

3.4.4.4 Pallet, lever, and escape wheel. The pallet, lever, and escape wheel shall be made of steel with a minimum carbon content of 0.75 percent and with a minimum degree of hardness equal to Rockwell C45.

3.4.5 Lever regulator. A lever regulator to adjust the clock time rate shall be accessible from the regulator opening on the case and shall be centered on the regulator scale, within 10 percent of either the "fast" or "slow" scale range.

3.4.6 Movement. The clock shall have an 8-day movement with not less than 15 jewels. End stone jewels shall be securely mounted in removable caps to facilitate cleaning and oiling. All jeweled bearings shall be in accordance with MIL-B-27497. Location of the jewels shall be as follows:

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Location	Number of jewels
Lever pallet	2
Roller	1
Balance (upper and lower pivot)	2 each
Pallet (upper and lower pivot)	1 each
Escape wheel (upper and lower pivot)	1 each
4th wheel (upper and lower pivot)	1 each
3rd wheel (upper and lower pivot)	1 each

3.4.6.1 Mainspring. The material for the mainspring shall be a corrosion-resistant, high-strength, unbreakable alloy.

3.4.7 Screw threads. Screw threads shall conform to MIL-S-7742.

3.4.8 Mounting screws and nuts.

3.4.8.1 Mounting screws for installing the clock shall be type 6-32UNC-2A roundhead, brass machine screws, having lusterless black oxidized or black nickel finish conforming to color No. 37038 of FED-STD 595. The screw length shall be sufficient to mount the clock on a 3/16-inch-thick panel.

3.4.8.2 Removable spring nuts, type 6-32UNC-2B, shall be furnished to fit each mounting lug.

3.4.8.3 A sufficient number of these screws and nuts shall be furnished in a marked envelope, size 2-1/4 by 4 inches, and packaged with each clock. The envelope shall be marked as follows:

IMPORTANT
This envelope contains mounting screws and nuts

3.4.9 Lighting. Lighting shall be unfiltered white only.

3.4.9.1 Lamps. Only lamps type 715AS15, 718AS15 and 328AS15 shall be used.

3.4.9.2 Receptacle MS27034H1035P or equivalent will be used. The power and ground sides of the integral lighting circuit shall terminate at pins 1 and 2 respectively in the instrument receptacle. Unless otherwise specified the ground side of the integral lighting circuit shall also be grounded to the instrument case with an internal grounding jumper having greater current carrying capacity than any other internal wire.

3.4.9.3 Location. The number of lamps and their location in the instrument shall be such that the intensity of any white lighted portion of the instrument shall not be reduced below 0.1 foot-Lambert at 2.70 $\pm$ .03 volts by the loss of any one lamp.

3.4.9.4 Intensity. The intensity of the white lighted portions of the instrument shall be .50 $\pm$ .25 foot-Lamberts at 2.70 $\pm$ .03 volts except on pointers where the brightness shall be .75 $\pm$ .25 foot Lamberts at 2.70 $\pm$ .03 volts. At 2.70 $\pm$ .03 volts the brightness of the black background shall not exceed 0.04 foot-Lamberts on non-transilluminated instruments and .0025 foot-Lamberts on transilluminated instruments.

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3.4.9.5 Glare. The lighting system shall be designed to shield all lamp filaments or glare sources from direct view, when the instrument is viewed from an angle of 60° either side of and above its centerline and from 30° below its centerline.

3.4.10 Reliability. The clock shall be designed and constructed to provide maximum practical periods of time between failures of the clock. The clock shall have a mean-time-between-failure (MT.) of at least 2,500 hours.

3.4.11 Maintenance provisions. The design of the clock shall permit ease and speed of maintainability for both preventive and corrective maintenance.

3.4.12 Lubrication. The clock shall be sufficiently lubricated for at least 2 years' operation. The lubricant shall be of such quality that satisfactory lubrication will be provided when the clock is run at temperatures of +55° to -35° C (+131° to 31° F).

3.5 Performance. The performance of the clock shall be as specified herein and shall show no deterioration as a result of the tests specified in section 4.

3.5.1 Elapsed time function operation. When the elapsed time knob is pushed either easy or hard, it shall operate one of the three phases of the elapsed time functions specified in 3.4.3.2. The triangle under the number "60" shall be the zero position. On the fly-back or zero push, both the sweep-second and the elapsed time hands shall return to the zero triangle and shall remain at this position until the knob is pushed for the starting function. The elapsed time hand shall rotate 360° in 12 minutes.

3.5.2 Rundown time. After being fully wound, the clock shall operate for a minimum of 8 days (192 hours) before stopping.

3.5.3 Rate and accuracy. (See 6.3).

3.5.3.1 Room temperature rate.

3.5.3.1.1 Daily rate (DR). (See 6.3). The average of the daily rates for 4 consecutive days shall not exceed 30 seconds.

3.5.3.1.2 Accuracy criterion. Any daily rate noted in the 4-day period shall not differ from the average daily rate (ADR) (see 6.3) by more than 15 seconds.

3.5.3.2 Temperature compensation rates.

3.5.3.2.1 Zero° C compensation. The rate of the clock when operated in an environment of 0° C for 6 hours shall not vary from the starting error (see 6.3) by more than 10 seconds.

3.5.3.2.2 Minus 35° C compensation. The rate of the clock when operated in an environment of -35° C for 6 hours shall not vary from the starting error by more than 75 seconds.

3.5.3.2.3 Plus 55° C compensation. The rate of the clock when operated in an environment of +55° C for 6 hours shall not vary from the starting error by more than 15 seconds.

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3.5.3.3 Vibration error rate. The rate of the clock when subjected to vibration as specified in 4.8.7 shall not vary from the starting error by more than 15 seconds during 4-1/2 hours. During vibration, the oscillation at the tip of the minute and second hands shall not exceed 0.0625 inch. No screws or parts shall be loosened nor any parts damaged as a result of the vibration error test.

3.5.3.4 Rates after environmental exposure. The rate of the clock for 6 hours, starting no later than 1 hour after exposure to each of the following environmental conditions, shall not vary from the starting error by more than the number of seconds specified below:

a. Temperature ranging from +71° to -62° C, nonoperating, and +55° to -35° C operating: 15 seconds.

b. Relative humidity up to 100 percent: 20 seconds.

c. Shock of 15g to be attained within 11 milliseconds ±10 percent: 20 seconds.

d. Vibration up to 2g within the range of 5 to 500 cycles per second (cps): 20 seconds.

e. Magnetic field intensity of 5.0 oersteds maximum: 15 seconds or 5 seconds, as applicable.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The identification and part numbering requirements of DOD-D-1000 shall govern the manufacturer's part numbers and changes thereto.

3.7 Weight. The weight of the clock shall not exceed 16 ounces.

3.8 Finishes. Aluminum-alloy parts shall be covered with an anodic film conforming to MIL-A-8625. Aluminum-alloy dials and small holes and parts on which anodic film would interfere with proper performance need not be anodized.

3.9 Operation markings.

3.9.1 Dial scale. The dial scale numerals shall distinctly indicate the graduation to which each applies.

3.9.2 Dial markings. The dial shall be marked in accordance with figure 2. The form of letters and numerals on the dial shall conform to MS33558. All markings shall be durable to withstand usage encountered in service.

3.9.3 Clock case marking. The clock case shall be clearly marked adjacent to the lighting receptacle as follow:

Wiring: 1 and 2 = White

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3.10 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.10.1 The clock movement shall be permanently stamped, engraved, or etched with the following information:

Manufacturer's name
Manufacturer's trademark or code
Manufacturer's movement serial number
Number of jewels
Compensated or uncompensated, as applicable

3.10.2 The clock case shall have a nameplate with the following information:

Clock, Aircraft, Mechanical ABU-32/A
Manufacturer's part No.
Manufacturer's serial No. (year)
Contract or order No.
Stock No. FSN 6645
Manufacturer's name:
US property

3.11 Workmanship. The aircraft clock, including all parts and accessories, shall be finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and marking of assemblies, and to ensuring that they are free from burrs and sharp edges.

3.11.1 Screw assemblies. Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further without damage to the screws, bolts, or threads.

3.11.2 Cleaning. The aircraft clock shall be thoroughly cleaned of metal chips and other foreign material at final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The examining and testing of the clock shall be classified as follows:

a. First article inspection (see 4.3).

b. Certification inspection (see 4.4).

c. Quality conformance inspection (see 4.5).

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4.3 First article inspections.

4.3.1 First article tests. First article tests shall consist of all tests specified in Table 1.

4.3.1.1 Data to accompany samples. The samples shall be accompanied by the following data:

a. Brief operating data to enable test personnel to correctly operate the clock.

b. Engineering data in the form of assembly drawings (two sets) and calibration record charts of test chambers, meter gages, and stands used to conduct qualification tests.

c. Certification that wiring conforms to 3.4.9.2 and sub-paragraphs thereto.

4.3.1.2 Testing schedule. The sample clocks shall be subjected to the tests as specified in Table I.

4.3.1.3 Reliability. For conformance to the reliability requirement of 3.4.10, the 10 samples shall pass the following tests when conducted as part of the first article tests specified in Table I.

Test	Applicable paragraph
Operation of elapsed time function	4.8.1
Rundown	4.8.2
Room temperature rate	4.8.3
Zero° C compensation	4.8.4
Minus 35° C compensation	4.8.5
Plus 55° C compensation	4.8.6
Vibration error	4.8.7
Lighting	4.8.9

4.3.2 Test report, test samples, and data. The contractor shall furnish the activity responsible for first article a report of the first article tests, including the following:

a. Test report: Three certified copies of the test report in accordance with MIL-STD-831.

b. Test samples: Two test samples of the clocks and two clocks which have passed individual tests (4.5.1).

c. Data: The data required under 4.3.1.1.

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TABLE I. First Article test schedule

Type of Test	Sample Number										Acceptable Paragraph
	1	2	3	4	5	6	7	8	9	10	
Examination of Product	X	X	X	X	X	X	X	X	X	X	4.7.1
Operation of elapsed function	X	X	X	X	X	X	X	X	X	X	4.8.1
Rundown	X	X	X	X	X	X	X	X	X	X	4.8.2
Room temperature rate	X	X	X	X	X	X	X	X	X	X	4.8.3
Zero° C compensation	X	X	X	X	X	X	X	X	X	X	4.8.4
Minus 35° C compensation	X	X	X	X	X	X	X	X	X	X	4.8.5
Plus 55° C compensation	X	X	X	X	X	X	X	X	X	X	4.8.6
Vibration error	X	X	X	X	X	X	X	X	X	X	4.8.7
High temperature	X	X									4.8.8.1
Low temperature			X	X							4.8.8.2
Humidity					X	X					4.8.8.3
Shock								X	X	X	4.8.8.4
Vibration									X	X	4.8.8.5
Operation in a magnetic field	X	X	X	X	X	X	X	X	X	X	4.8.8.6
Lighting	X	X	X	X	X	X	X	X	X	X	4.8.9

4.4 Certification inspection for primary standard. This procedure shall be a certification inspection procedure to establish a "lighting prime standard clock" to be used on production line testing only. Prior to acceptance testing, a production clock conforming to all requirements of the lighting specifications specified herein, along with certified lighting data on the clock shall be submitted for approval to the procuring activity. The clock shall be sealed prior to submission and should contain a mating connector with approximately 1 foot of low-impedance lead wire. A minimum of 12 average areas covering all representative areas on the dial of the clock will be measured for brightness and color as required in applicable specifications. If approved, the clock will be certified and identified as a "lighting prime standard" for a 90-day period.

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4.4.1 Effective life of prime standard. The certified "prime standard clock" will be used for visual comparison of production clocks for 100 operating hours or for 90 days, whichever comes first. If, during this period, a change of ± 20 percent in brightness or ± 5 percent in color takes place in two or more of the measured areas using the initially submitted data as a reference, the certification of the "prime standard clock" will be voided. (See 4.8.10.)

4.5 Quality conformance inspections. Quality conformance inspections shall consist of the examinations, individual tests, and sampling plans A and B tests.

4.5.1 Individual tests. Each clock shall be subjected to the following tests.

Tests	Applicable paragraph
Examination of product	4.7.1
Operation of elapsed time function	4.8.1
Room temperature rate	4.8.3
Visual lighting comparison	4.8.10
Reliability acceptance	4.8.11

4.5.2 Sampling plans A and B tests.

4.5.2.1 Sampling plan A. Unless otherwise specified by the procuring activity, 4 clocks shall be selected at random from the first 20 clocks of the initial production run and, having passed the individual tests (4.5.1), shall be subjected to the following tests:

Tests	Sample No. 1 (2 clocks)	Sample No. 2 (2 clocks)	Applicable Paragraph
High temperature	---	X	4.8.8.1
Low temperature	---	X	4.8.8.2
Humidity	X	---	4.8.8.3
Vibration	X	---	4.8.8.5
Operation in a magnetic field	X	---	4.8.8.6

4.5.2.2 Sampling plan B. Four clocks shall be selected at random from every 50, or fraction thereof, manufactured on the production run and shall be subjected to the following tests:

Tests	Applicable paragraph
Rundown	4.8.2
Zero° C compensation	4.8.4
Minus 35° C compensation	4.8.5
Plus 55° C compensation	4.8.6
Vibration error	4.8.7
Shock	4.8.8.4
Lighting	4.8.9

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4.5.3 Rejection and retest. When one clock selected from the production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.5.3.1 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of the clocks on hand or later produced shall not be made until it is determined that all clocks meet all the requirements of this specification.

4.5.3.2 Defects in clocks already accepted. The investigation of a test failure could indicate that defects may exist in clocks already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test conditions.

4.6.1 Standard atmospheric conditions. Unless otherwise specified, all tests shall be made at standard atmospheric pressure (approximately 29.92 inches of mercury) and at room temperature (approximately 25° C (77° F)). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the change in instrument reading.

4.6.2 Attitude. Unless otherwise specified, the clock shall be tested in its normal operating position with the face vertical.

4.6.3 Lighting. All tests except the 0° C compensation test (4.8.4), the -35° C compensation test (4.8.5), and the low-temperature test (4.8.8.2) shall be performed with the lighting circuit activated and the bulbs lit.

4.6.4 Temperature test precaution. To prevent condensation of moisture on the mechanism following exposure to the specified temperatures in the 0° C compensation test (4.8.4) and the -35° C compensation test (4.8.5) the clock shall remain in the closed test chamber until the temperature has returned to normal.

4.6.5 Vibration stand. A vibration stand shall be used that will vibrate at any desired frequency between 500 and 3,000 cycles per minute (cpm) and will so subject the instrument to vibration that a point on the instrument case will describe, in a plane inclined 45 degrees to the horizontal, a circle as specified in 4.8.7.

4.7 Examinations.

4.7.1 Examination of product. The clock shall be inspected to verify that the materials, design, and construction, necessary mechanical measurements, marking, and workmanship comply with this specification.

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4.7.2 Examination of preparation for delivery. Preparation for delivery shall be examined for conformance to section 5.

4.8 Test methods.

4.8.1 Operation of elapsed time function. The clock shall be fully wound and tested for action of the sweep second and elapsed time hands on start, stop, and flyback operation.

4.8.1.1 Elapsed time flyback (3-minute interval). The clock shall be tested for 4 successive operations of the sweep second and elapsed time hands over intervals of approximately 3 minutes, starting from zero (60) index. Upon operation of the flyback mechanism, the hands shall return to the exact index position.

4.8.1.2 Elapsed time flyback (6-minute interval). The clock shall be tested for 8 successive operations of the sweep second and elapsed time hands over intervals of approximately 6 minutes, starting from zero (60) index. Upon operation of the flyback mechanism, the hands shall return to the exact index position.

4.8.2 Rundown. The clock shall be fully wound and allowed to run down. It shall not run down before 8 days (192 hours). The actual running time shall be recorded.

4.8.2.1 This test may be run in conjunction with the room temperature rate test specified in 4.8.3.

4.8.3 Room temperature rate. The clock shall run for at least 1 day to break in the parts. The clock shall then be fully wound and the starting error recorded. The room temperature rate test shall continue for 4 consecutive days with daily readings and without rewinding. The elapsed time function shall be running continually. The daily rate, average daily rate, and the variation from the average daily rate shall be computed from the test data (see 6.3.3).

4.8.3.1 Average daily rate. The average daily rate for the consecutive 4-day test period shall not exceed 30 seconds.

4.8.3.2 Accuracy verification. The difference between any daily rate and the average daily rate shall not exceed 15 seconds.

4.8.4 Zero° C compensation. The clock shall be fully wound and then subjected to a temperature of 0° ±2° C (32° ±3.6° F) for a period of 6 hours. The clock shall not gain nor lose more than 10 seconds during this period.

4.8.5 Minus 35° C compensation. The clock shall be fully wound and then subjected to a temperature of -35° ±2° C (-31° ±3.6° F) for a period of 6 hours. The clock shall not gain nor lose more than 75 seconds during this period.

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4.8.6 Plus 55° C compensation. The clock shall be fully wound and then subjected to a temperature of +55° ±2° C (131° ±3.6° F) for a period of 6 hours. The clock shall not gain nor lose more than 15 seconds during this period.

4.8.7 Vibration error. The clock shall be fully wound and mounted and in its normal upright position on a vibration stand with all hands operating. The clock shall be vibrated for 30 minutes at each of the following frequencies 500, 750, 1,000, 1,250, 1,500, 1,750, 2,000, 2,500, and 3,000 cpm, so that a point on the clock describes a circle, the diameter of which shall be 0.009 to 0.011 inch. The oscillations of the hands shall not exceed 0.0625 inch. The clock shall not gain nor lose more than 15 seconds during this period. After the vibration error test, the clock shall be carefully examined for loosened screws or other parts and damage that would affect subsequent operation.

4.8.8 Environmental tests. The equipment shall be subjected to the following environmental tests in accordance with the methods of MIL-STD-810, except for the magnetic field test which shall be conducted as specified herein. The procedures of MIL-STD-810 shall be used with exceptions and additions indicated. It shall not be necessary to operate the clock at ambient conditions prior to conducting the environmental tests in order to collect comparison data for the "pretest performance record" as specified in MIL-STD-810. The required data may be collected in conjunction with the room temperature rate test (4.8.3) and used for comparison in checking clock performance after exposure to environmental limits.

4.8.8.1 High temperature. The clock shall be subjected to a high-temperature in accordance with Method 501, Procedure I, except that exposure to a temperature of 71° C shall be for 24 hours and the test specified in 4.8.1.1 shall be substituted for the phase of the test calling for operation at highest operating temperature.

4.8.8.1.1 Rate check. After completion (within 1 hour) of the 24-hour high-temperature exposure test, the clock shall be fully wound and operated for 6 consecutive hours at room temperature. The clock shall not gain nor lose more than 15 seconds during this 6-hour period.

4.8.8.2 Low temperature. The clock shall be subjected to a low-temperature test in accordance with Method 502, Procedure I, except that exposure to a temperature of -62° C shall be for 24 hours and the test specified in 4.8.8.2.1 shall be substituted for the phase of the test calling for operation at lowest operating temperature.

4.8.8.2.1 Rate check. After completion (within 1 hour) of the low-temperature exposure test, the clock shall be fully wound and operated for 6 consecutive hours at room temperature. The clock shall not gain nor lose more than 15 seconds during this 6-hour period.

4.8.8.3 Humidity. The clock shall be subjected to a humidity test in accordance with Method 507, Procedure I, except that the total time shall be 120 hours (5 cycles).

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4.8.8.3.1 Rate check. Within 1 hour after completion of the humidity test, the clock shall be fully wound and operated for 6 consecutive hours. The clock shall not gain nor lose more than 20 seconds during this 6-hour period.

4.8.8.4 Shock. The clock shall be subjected to a shock test in accordance with Method 516, Procedure I. An acceleration of 15g shall be attained with the clock operating and being oriented in each of the following positions:

a. Face up

b. Face down

c. The numeral "15" up

d. The numeral "30" up

e. The numeral "45" up

f. The numeral "60" up

4.8.8.4.1 Rate check. Within 1 hour after completion of the shock test, the clock shall be fully wound and operated for 6 consecutive hours. The clock shall not gain nor lose more than 20 seconds during this 6-hour period.

4.8.8.5 Vibration. The clock shall be subjected to a vibration test in accordance with Method 514, table 514-I title "Vibration Test Selection Chart" test data: 1CIA. The tests shall be performed in accordance with Procedure I of Method 514, except that no random vibration testing shall be required. All clock hands shall be operating during the test.

4.8.8.5.1 Rate check. Within 1 hour after completion of the vibration test, the clock shall be fully wound and operated for 6 consecutive hours. The clock shall not gain nor lose more than 20 seconds during this 6-hour period.

4.8.8.6 Operation in a magnetic field. The clock shall be subjected to the following field strengths for at least 10 seconds in the order given; and the rates shall be noted:

a. In a magnetic field with a horizontal strength of 0.15 to 0.22 oersted.

b. In a magnetic field with a horizontal strength of approximately 5.0 oersteds.

c. In a magnetic field with a horizontal strength of 0.15 to 0.22 oersted.

4.8.8.6.1 Rate check. Within 1 hour after each field strength subjection, the clock shall be fully wound and operated for 6 successive hours and the rates noted. The rates noted from b. and c. of 4.8.8.6 shall not differ by more than 15 seconds, and the rates noted from a. and c. shall not differ by more than 5 seconds.

4.8.9 Lighting. The clock shall be tested to comply with paragraph 3.4.9.

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4.8.10 Visual lighting comparisons. The clock shall be tested by comparing it with the "lighting prime standard clock" (see 4.4). Critical areas on the clock dial shall be compared for brightness and color with corresponding areas on the "prime standard clock" and shall conform to the "prime standard clock". The "prime standard clock" shall be energized with 2.7 +0.05 volts. The voltage on the colck under test shall be varied untill the brightness appears comparable to the "Prime standard clock". The measured voltage to the clock under test shall be between 2.55 and 2.85 volts.

4.8.10.1 If there is no approved "lighting prime standard clock", the clock shall be subjected to testing as specified in 4.8.9 in lieu of the visual lighting comparison test.

4.8.11 Reliability acceptance test. The reliability acceptance criterion specified in this paragraph shall be applied to each production run. This criterion shall be that no clocks shall fail to pass the room temperature rate test specified in 4.8.3.

5. PACKAGING

5.1 PRESERVATION. Preservation shall be Level A, C or Commercial as specified (see 6.2).

5.1.1 Level A. Each clock shall be preserved and packaged in accordance with the requirements of PPP-T-360, Group 3.

5.1.2 Level C. The Level C preservation and packaging of each clock shall conform to the MIL-STD-794 requirements for this level.

5.1.3 Commercial. The commercial preservation and packaging of each clock shall be in accordance with the requirements of ASTM D 3951.

5.2 PACKING. Packing shall be Level A, B, C, or Commercial, as specified (see 6.2).

5.2.1 Level A. Clocks shall be packed in overseas type containers conforming to PPP-B-601. Where practicable, the exterior shipping containers shall be of uniform shape and size and of minimum cube and tare.

5.2.2 Level B. Clocks shall be packed in weather resistant shipping containers conforming to PPP-B-636. The grade of each shipping container, determined by the gross weight and size, shall be limited to Special Requirements Table of the specification. Closure shall be in accordance with the container specification.

5.2.3 Level C. The Level C packing for clocks shall conform to the MILSTD-794 requirements for this level.

5.2.4 COMMERCIAL. The preserved clocks shall be packed in accordance with the requirements of ASTM D 3951.

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5.3 Palletization. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be palletized in accordance with MIL-STD-147. Palletized loads shall be uniform in size and quantities to the greatest extent possible. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking.

5.4.1 Levels A, B and C. In addition to any special or other identification markings required by the contract (see 6.2) each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129 and where applicable PPP-T-360.

5.4.2 Commercial. Commercial marking shall be in accordance with the requirements of ASTM D 3951.

6. NOTES

6.1 Intended use. The clocks are intended for use as the primary timepiece with an elapsed time capability in aircraft requiring an integrally lighted clock.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Stock No.

c. Levels of packaging and packing desired (see 5.1).

6.3 Definitions. For the purpose of this specification, the following definitions apply.

6.3.1 Error. Error is the algebraic time difference in seconds between the test clock and National Bureau of Standards broadcast time. (Station WWV.)

6.3.1.1 Starting error. Starting error is the error at start of test period.

6.3.2 Rate. Rate is the difference between the error at the start and the error at the end of a given time interval.

6.3.2.1 Daily rate. Daily rate is the rate in a 24-hour interval.

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6.3.3 The following is an example of rate and accuracy computation (measurements in seconds):

	Daily Reading	Daily Rate	Difference between each DR and the ADR
Starting error	0		
1st day	+30	+30	30: variation from 20=10
2nd day	+50	+20	20: no variation
3rd day	+60	+10	10: variation from 20=10
4th day	+80	+20	20: no variation
		80	Therefore, no DR differs
		80 ÷ 4= 20	from the ADR by more than
		20= ADR	15 seconds

6.4 First article. When a first article inspection is required, the item will be tested and should be a first article sample or it may be a standard production item from the contractor's current inventory. The first article shall consist of ten units. The first article samples shall be marked with the manufacturer's own part number, and the contracting officer should include specific instructions for examinations, tests, and approval of the first article.

6.5 International standardization agreement. Certain provisions of 3.4.1, 3.4.2, and 3.9.2 of this specification are the subject of international standardization agreements STANAG 3405 and ABC AIR STD 10/32. When amendment, revision, or cancellation of this document is proposed which will modify the international agreement concerned, the preparing activity takes appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accomodations.

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(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-87975	CLOCK, AIRCRAFT, MECHANICAL ABU-32/A

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* - Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* - Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* - Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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